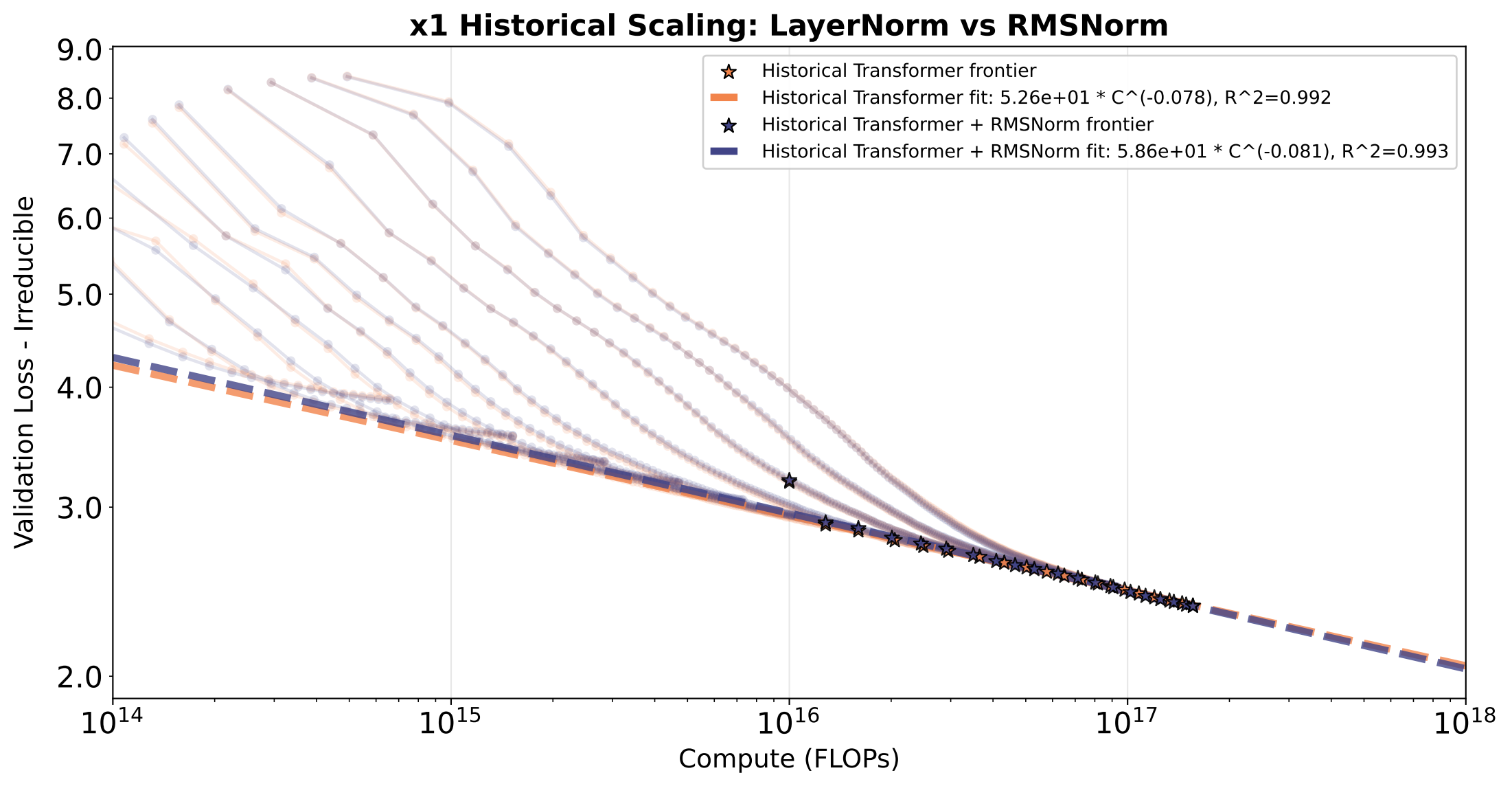

x1 Historical Scaling: LayerNorm vs RMSNorm
Historical Transformer frontier
Historical Transformer fit: 5.26e+01 * C^(-0.078), R^2=0.992
Historical Transformer + RMSNorm frontier
Historical Transformer + RMSNorm fit: 5.86e+01 * C^(-0.081), R^2=0.993
Validation Loss - Irreducible
9.0
8.0
7.0
6.0
5.0
4.0
3.0
2.0
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}
Compute (FLOPs)